

# BYUNGWOO JEON

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## RESEARCH INTEREST

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The north star of my research is to develop a unified and scalable framework aligned with multi-modalities and human preferences for various vision-centric tasks. To this end, I am delving into multi-modal learning (e.g., VLM, UMM) and robot learning (e.g., VLA), with a focus on 3D/4D world modeling that bridges perception, cognition and action for real-world embodied intelligence.

- **Multi-modal learning**
  - Reasoning
  - Omni-modal generation
  - World modeling
- **High-dimensional vision**
  - Spatio-temporal understanding
  - Motion & pose estimation
- **Robot learning**
  - Long-horizon planning
  - World action modeling

## PUBLICATIONS

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C: conference, J: journal, W: workshop, P: preprint / \* equal contribution, † equal advising

- [C3] **SpatialBoost: Enhancing Visual Representation through Language-Guided Reasoning**  
Byungwoo Jeon<sup>\*</sup>, Dongyoung Kim<sup>\*</sup>, Huiwon Jang, Insoo Kim, Jinwoo Shin  
*European Conference on Computer Vision (ECCV), 2026*
- [C2] **Vision-aligned Latent Reasoning for Multi-modal Large Language Model**  
Byungwoo Jeon, Yoonwoo Jeong, Hyunseok Lee, Minsu Cho<sup>†</sup>, Jinwoo Shin<sup>†</sup>  
*International Conference on Machine Learning (ICML), 2026*
- [C1] **TrackIME: Enhanced Video Point Tracking via Instance Motion Estimation**  
Seong Hyeon Park, Huiwon Jang, Byungwoo Jeon, Sukmin Yun, Paul Hongsuck Seo, Jinwoo Shin  
*Conference on Neural Information Processing Systems (NeurIPS), 2024, **Spotlight Presentation (326/15671=2.08%)***
- [P4] **Latent Reasoning in Continuous Space for Unified Multimodal Models**  
Byungwoo Jeon, Yoonwoo Jeong, Hyunseok Lee, Minsu Cho<sup>†</sup>, Jinwoo Shin<sup>†</sup>  
*Under review*
- [P3] **Physically Plausible Multi-View Object Swapping for Robot Data Augmentation**  
Jonghoon Lee<sup>\*</sup>, Seong Hyeon Park<sup>\*</sup>, Byungwoo Jeon, Minha Lee, Jinwoo Shin  
*Under review*
- [P2] **Cog3DMap: Multi-View Vision-Language Reasoning with 3D Cognitive Maps**  
Chanyoung Gwak<sup>\*</sup>, Yoonwoo Jeong<sup>\*</sup>, Byungwoo Jeon, Hyunseok Lee, Jinwoo Shin<sup>†</sup>, Minsu Cho<sup>†</sup>  
*Under review*
- [P1] **Visual Tokenization with Hierarchical Abstraction for Unified Multimodal Modeling**  
JB Kim, Hyeonmo Kang, Byungwoo Jeon, June Suk Choi, Kyungmin Lee, Jinwoo Shin<sup>†</sup>, Jeonghoon Kim<sup>†</sup>  
*Under review*
- [W1] **Hierarchical Vision Language Action Model Using Success and Failure Demonstrations**  
Jeongeun Park, Jihwan Yoon, Byungwoo Jeon, Juhan Park, Jinwoo Shin, Kyungjae Lee, Namhoon Cho, Sangdoo Yun, Sungjoon Choi  
*CoRL Workshop on Safe and Robust Robot Learning for Operation in the Real World, 2025*

## EDUCATION

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|------------------------------------------------------------------------------|---------------------|
| KAIST, Ph.D. student in Artificial Intelligence (advisor: Prof. Jinwoo Shin) | Sep 2024 - Current  |
| Korea University, B.S. in Computer Science and Engineering, Statistics       | Mar 2020 - Aug 2024 |

## WORK EXPERIENCE

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|---------------------------------------------------------|---------------------|
| ALINLAB, Research Intern (advisor: Prof. Jinwoo Shin)   | Mar 2023 - Aug 2024 |
| Arcreal, ML Engineer (host: Prof. Jinwoo Shin)          | Jan 2023 - Jan 2024 |
| MLVLAB, Research Intern (advisor: Prof. Hyunwoo J. Kim) | Jul 2022 - Dec 2022 |
| M-monstar, Fullstack Engineer                           | Jul 2021 - Aug 2021 |

## EXTERNAL COLLABORATOR

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| POSTECH CVLAB, Working on VLM reasoning [C2, P4, P2] (advisor: Prof. Minsu Cho) | Apr 2025 - Current |
| Korea Univ. RILAB, Working on VLA reasoning [W1] (advisor: Prof. Sungjoon Choi) | Jul 2025 - Current |
| ETH-Zürich CVG, Working on 4D world modeling (advisor: Dr. Sunghwan Hong)       | Jul 2026 - Current |

## PROJECTS

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### *Industry Projects*

NVIDIA, Latent World Models (2026–)  
RLWRLD, RLDX: Vision-Language-Action Model for Dexterous Manipulation (2026–)  
SAMSUNG, Semiconductor Pattern Generation (2025–2027)

### *Personal Projects*

Google Developer Groups, Google Solution Challenge: Recovery (2024), digiHow (2023)  
Google Developer Groups, Horang Studio: Diffusion-based personalized AI profile service (2023)

## ACADEMIC SERVICES

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Conference Reviewer: ECCV'26  
Workshop Reviewer: Space in Vision, Language, and Embodied AI (SpaVLE) @ NeurIPS'25

## HONORS & AWARDS

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Silver Prize, K-Data Science Hackathon, 2023 (\$2,000)  
Top 8, Artificial Intelligence Grand Challenge, 2022  
Dean's List, Korea University, 2020–2023

## EXTRACURRICULAR ACTIVITIES

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|--------------------------------------------------------------------------------|---------------------|
| Google Developer Groups on Campus (GDGoC) @ Korea University, Lead and Founder | Aug 2022 - Jul 2023 |
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## SELECTED TALKS

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Korea Univ., GDGoC, "Meta Runner on AI," Mar 2026  
Yonsei & Korea Univ., GDGoC, "Towards the New World of Computing: CV on Real-world," Dec 2024  
Korea Univ., GDGoC, "Recent Advances in Machine Learning," Nov 2023

## MENTORING

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- **Byungjun Yoon**, Ph.D. student at KAIST AI  
Topic: Hierarchical VLA  
Mar 2025 - Current
- **Jonghoon Lee**, M.S. student at KAIST AI  
Topic: 3D/4D mesh generation [P4]  
Apr 2025 - Current
- **Minha Lee**, M.S. student at KAIST AI  
Topic: 3D/4D mesh generation [P4], Hierarchical world modeling  
Mar 2026 - Current
- **Kiyoung Kwon**, now senior undergrad at SNU AI  
Topic: Agentic AI4Science  
Jan 2026 - Current
- **Hyunseo Yang**, now senior undergrad at SNU CSE  
Topic: GUI agent with world modeling  
Jul 2026 - Current
- **Heejun Jung**, now senior undergrad at KAIST SoC  
Topic: Scalability of world modeling  
Jul 2026 - Current
- **Seoyeon Byun**, M.S. student at KAIST AI  
Topic: 3D/4D neural rendering, **Collaborate**: GDGoC 3rd Lead  
Aug 2022 - Aug 2024
- **Seonghu Jeon**, M.S. student at KAIST AI  
Topic: 3D/4D neural rendering, **Collaborate**: GDGoC 3rd AI Core  
Mar 2022 - Aug 2024

## REFERENCE

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**Jinwoo Shin**, Full Professor at KAIST

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**Minsu Cho**, Full Professor at POSTECH

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**Sungjoon Choi**, Assistant Professor at Korea Univ.

E-mail: [sungjoon-choi@korea.ac.kr](mailto:sungjoon-choi@korea.ac.kr)

**Seong Hyeon Park**, incoming Assistant Professor at Ajou University

E-mail: [seonghyp@kaist.ac.kr](mailto:seonghyp@kaist.ac.kr)

**Sunghwan Hong**, Postdoc fellow at ETH-Zürich

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